

STUDENT NUMBER: _____



YEAR 11 MATHEMATICS

September 2011

PRELIMINARY EXAMINATION

Time allowed: 2 hours plus 5 minutes reading time

INSTRUCTIONS

- This examination paper contains two sections with a total value of 65 marks.

Section I: Objective Response Questions (5 questions of 1 mark each)
Answer on the Objective-Response Answer Sheet provided.

Section II: Extended Response Questions (4 questions of 15 marks each)
Start each question on a new page.

- A table of standard integrals is provided.
- Board-approved calculators may be used.
- Geometric equipment and mathematical curve-drawing templates are allowed.
- Marks may not be awarded for untidy or careless work.
- More marks will be awarded for questions involving higher-order thinking.
- You may tear off the Objective-Response Answer Sheet.

SECTION I : Objective Response Questions**5 Marks***Answer on the Objective-Response Answer Sheet provided.*

1. The equation of the line through the point (9, 7) and parallel to the x -axis is
- (A) $x = 9$ (B) $y = 7$
(C) $y = 9$ (D) $x = 7$
2. The function $f(x) = x^2 - 2x$ is
- (A) even (B) odd
(C) neither (D) positive definite
3. The gradient of a line perpendicular to $3x - 4y + 7 = 0$ is
- (A) $\frac{3}{4}$ (B) $\frac{-3}{4}$
(C) $\frac{4}{3}$ (D) $\frac{-4}{3}$
4. $81^{-\frac{3}{4}}$ is **not** equal to
- (A) $\frac{1}{81^{\frac{3}{4}}}$ (B) $\frac{1}{27}$
(C) $\frac{1}{27^4}$ (D) $0.\dot{0}3\dot{7}$
5. Express the equation of the parabola $x^2 = 2(2y - 1)$ in the form $(x - h)^2 = 4a(y - k)$ and hence write down the focal length.

SECTION II: Extended Response Questions**60 Marks***Start each question on a new page.*

Question 1 (*Start a new page.*)**15 Marks**

- (a) Evaluate, correct to three significant figures [2]

$$\frac{13^5}{9^6 + 11^4} + 1$$

- (b) Factorise $16x^2 - 25$. [2]

- (c) Solve $2x + 3 \leq 8$. [2]

- (d) Find all the value(s) of x in the interval $0^\circ \leq x \leq 360^\circ$ for which $\tan x = \frac{1}{\sqrt{3}}$. [2]

- (e) For what value(s) of k does $3x^2 + 2x + k = 0$ have real roots? [2]

- (f) Simplify $\frac{5}{x^2 + x} + \frac{2}{x^2 - 1}$. [3]

- (g) Simplify $\tan \theta \cos \theta$. [2]

Question 2 (*Start a new page.*)**15 Marks**

(a) Differentiate

(i) $x^2 - 3x$ [2]

(ii) $\frac{1}{x}$ [2]

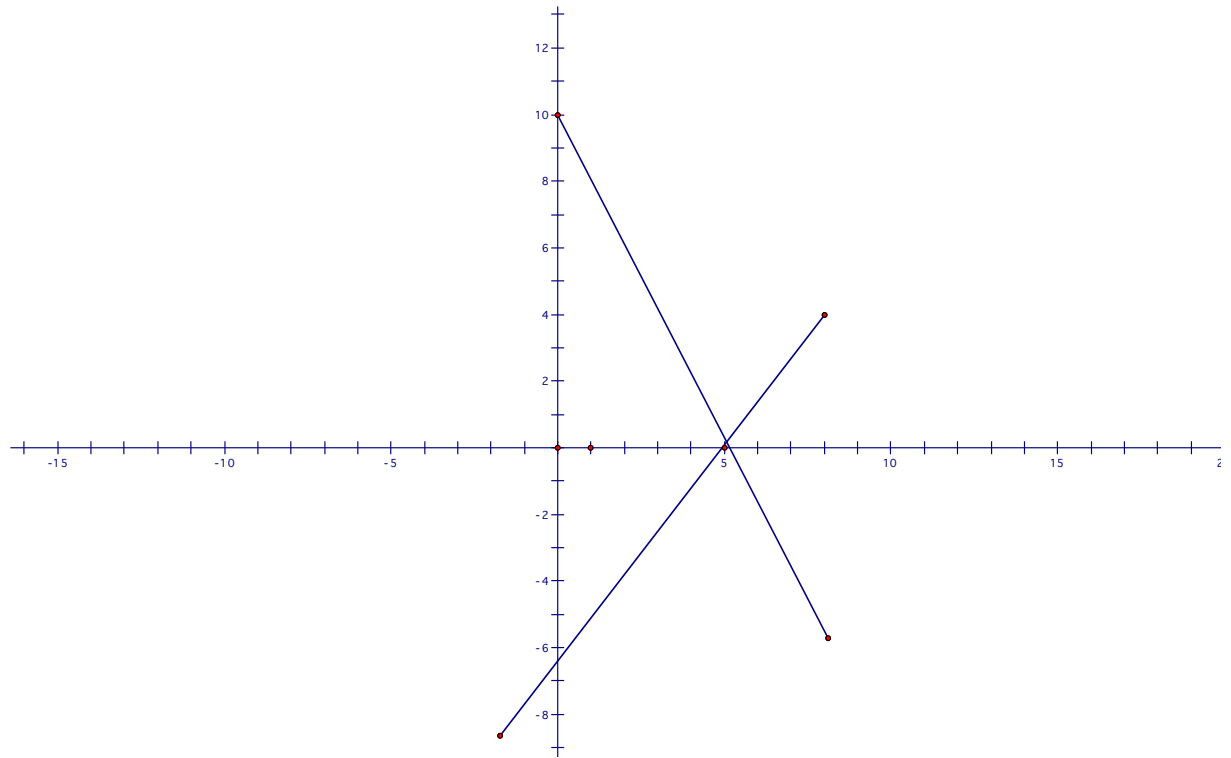
(b) (i) Sketch the graph of $y = |2x - 6|$, showing all relevant features. [2](ii) On the same set of axes, sketch the line $y = -1$. [1](iii) Hence state number of solutions to the equation $|2x - 6| = -1$. [1](c) Given $x^2 - (k + 5)x + 5k = 0$, find the value(s) of k such that one root is two more than the other. [3](d) A parabola has focus $(2, 1)$ and directrix $y = 5$.

(i) Find the vertex. [1]

(ii) Sketch the parabola and write down its equation. [3]

Question 3 (*Start a new page.*)**15 Marks**

- (a) If $f(x) = 1 - x^2$,
- (i) find $f(x + a)$ [1]
- (ii) find, in simplest form, $\frac{f(x + a) - f(x)}{a}$. [2]
- (b) (i) Show that $1 + \frac{2}{x} = \frac{x + 2}{x}$. [1]
- (ii) Find the domain and range of the curve $y = \frac{x + 2}{x}$. [2]
- (c) Given that the sides of a right-angled triangle are x , $x + 1$ and $x + 2$, find the value of x . [3]
- (d) In the diagram below, the point B(8, 4) lies on line L_1 and the point C(0, 10) lies on the line L_2 . The lines L_1 and L_2 meet at the point A(5, 0). The point M lies on the y-axis.

**Question 3 - Continued**

- (i) Show that the gradient of AB is $\frac{4}{3}$. [1]
- (ii) Find the angle that the line AB makes with the positive direction of the x -axis. [1]
- (iii) Show that the equation of the line AB is $3y = 4x - 20$. [1]
- (iv) The line AB cuts the y -axis at M. Find the co-ordinates of the point M. [1]
- (v) Find the area of ΔCMA to one decimal place. [2]

Question 4 (*Start a new page.*)**15 Marks**

- (a) Simplify $\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 2 \sec^2 \theta$. [2]

- (b) (i) Find the perpendicular distance between the line $3x + 4y - 3 = 0$ and the circle $x^2 + y^2 = 16$. [2]
- (ii) Is the line a tangent or secant to the circle, or neither? Give reasons for your answer. [1]
- (c) Write down a possible equation of a parabola that is negative definite. [1]
- (d) Given the points A (2, 4) and B (-4, 2), find the equation of the locus of the point P(x, y) and describe the locus geometrically if $\angle APB$ is a right-angle. [3]

Question 4 -Continued

- (e) In the diagram below, $AD = 2\text{cm}$, $OC = 3\text{ cm}$ and $\angle DAB = 30^\circ$. O is the centre of the circle.

- (i) Show that $\angle OCB = 56^\circ$ (to the nearest degree) [2]
- (ii) Find the area of triangle OCB. [2]

END OF EXAMINATION

Objective Response Answers

Student Number: _____

Tear off this sheet and hand it in separately.

Questions 1-4 : **Circle** the correct answer

Question 5: Write the correct answer in the box provided

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5				

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

SECTION I

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Qn	Solutions	Marks	Comments+Criteria
1.	parallel to x axis (B) $\therefore y = 7$		
2	$f(x) = \frac{x^2 - 2x}{}$ $f(-x) = (-x)^2 - 2x - x$ $= \frac{x^2 + 2x}{}$ $\therefore$ NEITHER (C)		
3.	m in general form $= -\frac{a}{b}$ $= -\frac{3}{4}$ $\therefore$ perpendicular $= -\frac{4}{3}$ (D)		
4	$81^{-3/4} = \frac{1}{81^{3/4}} = \frac{1}{\sqrt[4]{81^3}} = \frac{1}{3^3} = \frac{1}{27}$ NOT-C (C)		
5.	$x^2 = 2(2y-1)$ $= 4(y-\frac{1}{2})$ $x^2 = 4a(y-k)$ $4a = 4$ <u>$a = 1$</u>		

QUESTION 1

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Qn	Solutions	Marks	Comments+Criteria
(a)	$1.67 \overline{992}$ $= 1.68$	1 1	always write down your answer from calc before rounding. if wrong answer but correct rounding 1-2
(b)	$16x^2 - 25$ $(4x-5)(4x+5)$		
(c)	$2x + 3 \leq 8$ $2x \leq 8 - 3$ $2x \leq 5$ $x \leq 2\frac{1}{2}$	1 1	
(d)	$\tan x = \frac{1}{\sqrt{3}}$ positive ... Q1+3 acute angle $x = 30^\circ$ Q3 $180 + 30 = 210^\circ$ $x = 30^\circ$ or 210°	1 1	
(e)	$\Delta \geq 0$ $3x^2 + 2x + K = 0$ $b^2 - 4ac \geq 0$ $4 - 4 \times 3 \times K \geq 0$ $-12K \geq -4$ $K \leq \frac{-4}{-12}$ $K \leq \frac{1}{3}$		Real roots $\Delta \geq 0$ many wrote equal $\div$ by - change inequality sign

QUESTION 1 CONTINUED

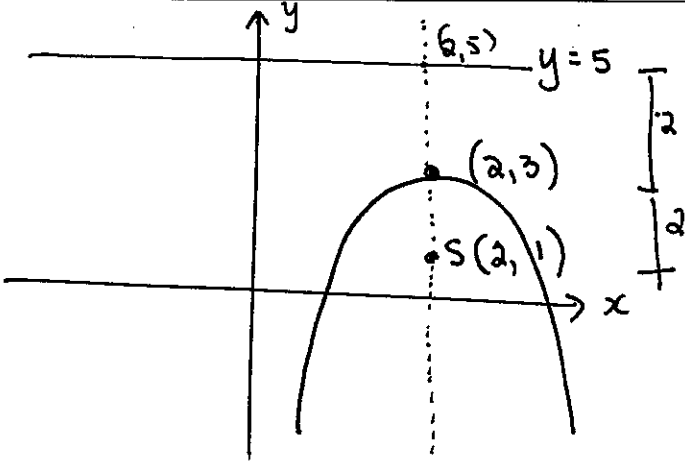
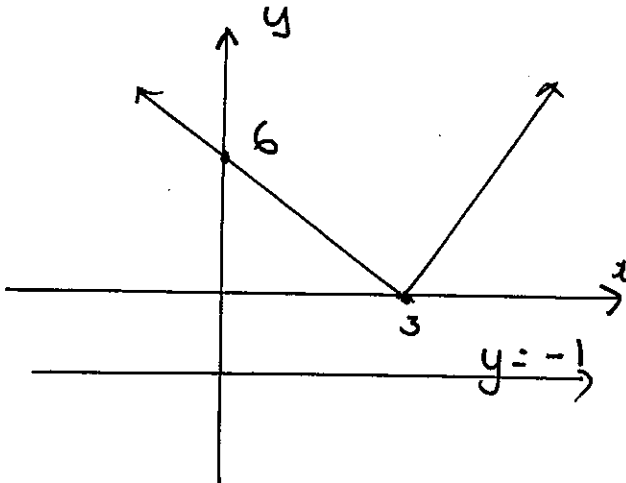
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Qn	Solutions	Marks	Comments+Criteria
(f)	$\frac{5}{x^2+x} + \frac{2}{x^2-1}$ $\frac{5}{x(x+1)} + \frac{2}{(x+1)(x-1)}$ $\frac{5(x-1) + 2x}{x(x+1)(x-1)}$ $\frac{5x - 5 + 2x}{x(x+1)(x-1)}$ $\frac{7x - 5}{x(x+1)(x-1)}$	max 1 if not factorised and wrong answer 1 1 1	Much easier to factorise Those who didn't often ended up with <u>NOT</u> the simplest fraction Could have different forms of the denominator
(g)	$\frac{\tan \theta \cdot \cos \theta}{\frac{\sin \theta}{\cos \theta} \cdot \cos \theta} = \sin \theta$	2	An easy 2 marks.

Q2

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Qn	Solutions	Marks	Comments+Criteria
(a)	(i) $y = x^2 - 3x$ $y' = 2x - 3$ (ii) $y = \frac{1}{x} = x^{-1}$ $y' = -1x^{-2}$ $= -\frac{1}{x^2}$	(2) 1 1 (2)	
(b)	$x^2 - (k+5)x + 5k = 0$ Let first root $= \alpha$ second root $= \alpha + 2$ $\alpha + \alpha + 2 = -\frac{b}{a}$ $2\alpha + 2 = k+5$ (1) $2\alpha = k+3$ $2\alpha - 3 = k$ $\alpha(\alpha+2) = \frac{c}{a} = 5k$ (2) $\alpha^2 + 2\alpha = 5k$ $\alpha^2 + 2\alpha = 5(2\alpha - 3)$ subst. $\alpha^2 + 2\alpha = 10\alpha - 15$ $\alpha^2 - 8\alpha + 15 = 0$ $(\alpha - 5)(\alpha - 3) = 0$ $\alpha = 3 \text{ or } 5$ $k = 2\alpha - 3$ or $2\alpha - 3$ $= 2 \times 5 - 3$ $2 \times 3 - 3$ $= 10 - 3$ $= 7$ $6 - 3$ <u> </u> <u> 3 </u>	1 for both eqn.	

Qn	Solutions	Marks	Comments+Criteria
(c) (i)	 Vertex $(2, 3)$ $a = -2$ $(x-2)^2 = -4a(y-3)$ $(x-2)^2 = -8(y-3)$		no penalty for inconsistent scale or for incorrect y intercept. 1. correct directrix 1. correct eqn 1. focal length
	 $2x - 6 = -1$ has no solutions.		

Q3

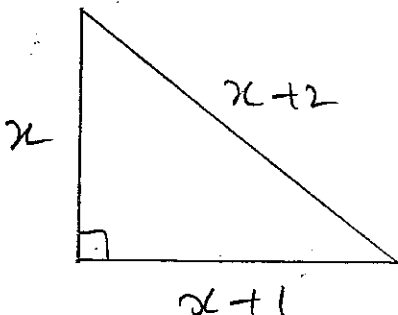
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Qn	Solutions	Marks	Comments+Criteria
(a)	C (10,6) A (5,0)		
(i)	$m = \frac{0-6}{5-10} = \frac{6}{5}$	1	
(ii)	$y-0 = \frac{6}{5}(x-5)$ $5y = 6x - 30$ $0 = 6x - 5y - 30$	1	
(iii)	cuts y axis $x=0$ $0 = 0 - 5y - 30$ $5y = -30 \therefore y = -6$ M (0,-6)	1	
(iv)	$A = \frac{1}{2}bh = \frac{1}{2} \times 16 \times 5$ $= 400^2$	1	
(v)	B (0,10) A (5,0) $m = \frac{0-10}{5-0} = -2$ $\therefore -ve \therefore$ obtuse angle acute angle $63^\circ 26' (n.m)$ obtuse $180 - 63^\circ 26'$ $116^\circ 34'$		

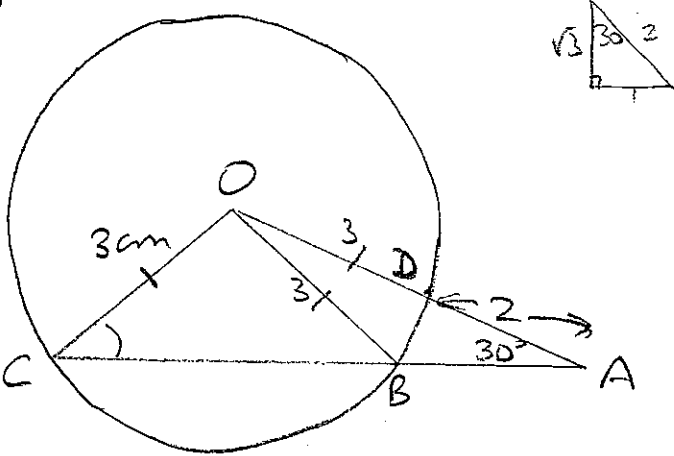
Q3

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Qn	Solutions	Marks	Comments+Criteria
(b)	$\text{LHS} = \frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta}$ $\frac{1 + \sin \theta + 1 - \sin \theta}{(1 - \sin \theta)(1 + \sin \theta)}$ $\frac{2}{1 - \sin^2 \theta} = \frac{2}{\cos^2 \theta} = 2 \sec^2 \theta = \text{RHS}$	1 1	
(c)	$f(x) = 1 - x^2$ $f(x+a) = 1 - (x+a)^2$ $= 1 - (x^2 + 2ax + a^2)$ $= 1 - x^2 - 2ax - a^2$ $\frac{f(x+a) - f(x)}{a} = \frac{1 - x^2 - 2ax - a^2 - 1 + x^2}{a}$ $= \frac{-2ax - a^2}{a}$ $= -(2x + a)$	1 1 1	
(d)	PA PB $\left(\frac{y-4}{x-2} \right) \cdot \left(\frac{y-2}{x+4} \right) = -1$ $\frac{y^2 - 6y + 8}{x^2 + 2x - 8} = -1$ $y^2 - 6y + 8 = -1(x^2 + 2x - 8)$ $y^2 - 6y + 8 = -x^2 - 2x + 8$ $x^2 + 2x + y^2 - 6y = 0$ $x^2 + 2x + 1 + y^2 - 6y + 9 = 1 + 9$ $(x+1)^2 + (y-3)^2 = 10$ Circle centre $(-1, 3)$ $r = \sqrt{10}$		

Qn	Solutions	Marks	Comments+Criteria
4	(a)  $(x+2)^2 = x^2 + (x+1)^2$ $\cancel{x^2} + 4x + 4 = x^2 + \cancel{x^2} + 2x + 1$ $0 = x^2 - 2x - 3$ $(x-3)(x+1) = 0$ $x = 3, x = -1.$ but $x > 0$, $\therefore x = 3$ only.	1 1 1	
	(b) (i) LHS = $1 + \frac{2}{x} = \frac{x}{x} + \frac{2}{x}$ $= \frac{x+2}{x}$ = RHS	1	proof
	(ii) $y = \frac{x+2}{x} = 1 + \frac{2}{x}$ D: all real x except $x=0$ R: all real y except $y=1$.	1 1	

[illegible]

Qn	Solutions	Marks	Comments+Criteria
4 old	(e)  (i) In $\triangle OCB$, $\frac{\sin 30}{3} = \frac{\sin(\angle OCB)}{5}$ $\sin(\angle OCB) = 5 \times \frac{1}{3}$ $\sin(\angle OCB) = \frac{5}{6}$ $\therefore \angle OCB = \sin^{-1}\left(\frac{5}{6}\right)$ $= 56^{\circ} 26' 33.68''$ $= 56^{\circ} \text{ (nearest deg.)}$	1 1	correct substitution into sine rule
	(ii) $\triangle OCB$ is isosceles (equal radii). $OC = OB = 3\text{cm}$ $\therefore$ base $\angle$s = . $\angle COB = 180 - 2 \times 56$ $= 68^{\circ}$ Area of $\triangle OCB = \frac{1}{2} ab \sin C$ $= \frac{1}{2} \times 3 \times 3 \times \sin 68^{\circ}$ $= 4.1723 \dots$	1 1	

$$= 4.17 \text{ cm}^2 \text{ (2dp's)}$$